

A Diagonal Exponent Improvement for the Matrix-Weighted Auxiliary Maximal Operator

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Source Question

Kakaroumpas and Soler i Gibert prove in arXiv:2407.16776, Theorem 6, that for a matrix A_p weight W and $1 < p, q < \infty$,

$$\left\| \left(\sum_{n=1}^{\infty} |\widetilde{M}_W f_n|^q \right)^{1/q} \right\|_{L^p(\mathbb{R}^N)} \leq C[W]_{A_p}^{\frac{1}{p} + \max\{\frac{1}{q-1}, \frac{1}{p-1}\}} \left\| \left(\sum_{n=1}^{\infty} |W^{1/p} f_n|^q \right)^{1/q} \right\|_{L^p(\mathbb{R}^N)}.$$

Immediately before the theorem they note that they do not know whether this exponent can be improved. The following observation gives such an improvement on the diagonal $q = p$.

Proof Intuition

The loss in Theorem 6 comes from a pointwise domination of $\widetilde{M}_W f$ by a Hardy–Littlewood maximal operator applied to the Christ–Goldberg maximal operator. On the diagonal $q = p$ no vector-valued machinery is needed: the $L^p(\ell^p)$ norm is exactly the sum of the individual L^p norms. Thus the known one-function estimate for $\widetilde{M}_W$ can be applied coordinate by coordinate.

Theorem 1. *Let $1 < p < \infty$, let W be a $(d \times d)$ matrix A_p weight on $\mathbb{R}^N$, and let (f_n) be a finite or countable sequence of locally integrable $\mathbb{C}^d$ -valued functions. Then*

$$\left\| \left(\sum_{n=1}^{\infty} |\widetilde{M}_W f_n|^p \right)^{1/p} \right\|_{L^p(\mathbb{R}^N)} \leq C_{N,d,p} [W]_{A_p}^{1/(p-1)} \left\| \left(\sum_{n=1}^{\infty} |W^{1/p} f_n|^p \right)^{1/p} \right\|_{L^p(\mathbb{R}^N)}.$$

Proof. The source paper recalls, just before its vector-valued estimates, the one-function bound

$$\|\widetilde{M}_W f\|_{L^p(\mathbb{R}^N)} \leq C_{N,d,p} [W]_{A_p}^{1/(p-1)} \| |W^{1/p} f| \|_{L^p(\mathbb{R}^N)},$$

originally due to Isralowitz–Kwon–Pott for the auxiliary matrix-weighted maximal operator.

For a finite sequence $(f_n)_{n=1}^m$, raise the desired estimate to the p th power and use the one-function estimate termwise:

$$\begin{aligned}
\left\| \left(\sum_{n=1}^m |\widetilde{M}_W f_n|^p \right)^{1/p} \right\|_{L^p}^p &= \sum_{n=1}^m \|\widetilde{M}_W f_n\|_{L^p}^p \\
&\leq C_{N,d,p}^p [W]_{A_p}^{p/(p-1)} \sum_{n=1}^m \| |W|^{1/p} f_n \|_{L^p}^p \\
&= C_{N,d,p}^p [W]_{A_p}^{p/(p-1)} \left\| \left(\sum_{n=1}^m |W|^{1/p} f_n|^p \right)^{1/p} \right\|_{L^p}^p.
\end{aligned}$$

Taking p th roots proves the finite-sequence case. The countable case follows by monotone convergence applied to the partial sums on both sides. $\square$

Remark 1. For $q = p$, Theorem 6 in the source gives the larger exponent

$$\frac{1}{p} + \max \left\{ \frac{1}{p-1}, \frac{1}{p-1} \right\} = \frac{1}{p} + \frac{1}{p-1}.$$

The theorem above removes the additional $1/p$ loss for all matrix A_p weights. It does not settle the off-diagonal vector-valued estimates $q \neq p$.

References

- S. Kakaroumpas and O. Soler i Gibert, *Vector valued estimates for matrix weighted maximal operators and product BMO*, arXiv:2407.16776.
- I. Isralowitz, K. Kwon, and S. Pott, one-function bound for the Christ–Goldberg auxiliary maximal operator, as cited in arXiv:2407.16776.